P1: Riemann–Roch system with a Dickson-torus tower on NIST-shape primes

research/split_prime_ecdlp_20261107/followups/p1_rr_dickson · run-20261107T214621Z · 2026-11-07

Outcome under the frozen rules: rule 5, stage-level positive at $m = 3$; rule 6 undecided [pcc:unverified]. On T24 = 14,680,063 and T31 = 2,046,820,351 ($p \equiv 3 \bmod 4$, $2^t \mid p+1$), the separator-free Riemann–Roch system (RR) carries the Dickson-torus tower with exactly its Kummer-tower solving degree and beats the cherry-tree chain (T) on degree and widest matrix at m3 t3 and m3 t4 on both primes. A transfer of round7_rr's Kummer result to the NIST field shape, not a new mechanism; a stage diagnostic (S, end-to-end cost, rho comparison unset). Card `CARD-p1-run-01`: executed as specified (director). Host: this container, L0.

1. Question and method

round7_rr found D_RR = 4 < D_T = 6 at m3 t4 with a Kummer tower [pcc:unverified]; NIST-shape primes admit only the Dickson torus tower. Protocol frozen 535d9da826 (Addendum A before any claim cell, merged in #3657). Instrument `p1tool` (Rust; emits round 7's systems byte for byte, reproduces round7_rr's D and widths exactly). msolve 0.6.5, 1.5 GiB, 1,200 s, one job at a time; 10 cells × 2 seeds × 2 systems = 40 jobs; analysis `p1analyze` (Rust).

2. Results (max over 2 seeds; "≥" = capped run, a lower bound)

cell	D RR	D T	widest RR	widest T	status
T24 m3 t3	4	5	632	883	both completed
T24 m3 t4	4	6	2,801	7,129	both completed
T24 m3 t5	5	≥ 6	31,911	≥ 23,852	T capped
T24 m5 t3	5	≥ 8	2,878	≥ 39,898	T capped
T31 m3 t3	4	5	632	883	both completed
T31 m3 t4	4	6	2,801	7,129	both completed
T31 m3 t5	5	≥ 6	31,911	≥ 23,852	T capped
p0 Kummer m3 t3	4	5	632	883	both completed
p0 Kummer m3 t4	4	6	2,801	7,077	both completed
p0 Kummer m3 t5	5	≥ 6	31,870	≥ 23,720	T capped

Frozen rules: 1 control met (RR-Kummer m3 t4 D = 4; every completed job finds the planted relation); 2 not killed; 3 transfer holds (D_RR Dickson = Kummer at every shared cell; widths within 0.2 %); 4 no trade in completed cells; **5 yes** (m3 t3, m3 t4, both primes); **6 undecided** (T capped at m5 t3; the frozen text does not decide a one-arm cap). A post-data reading of capped values as lower bounds (sound) would give "met"; it is labelled and not reported as the outcome.

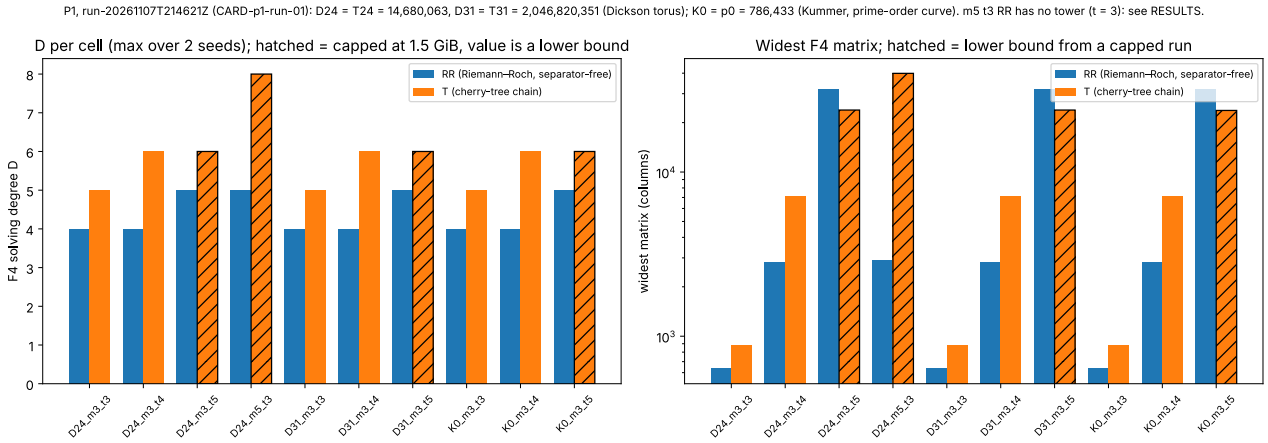


Figure 1: D and widest matrix per cell; hatched bars are capped runs (lower bounds). From run-20261107T214621Z via `p1analyze`.

3. Red-team findings that bound the reading

(1) At m5 t3 the RR system has no tower: with $t = 3$ the membership is one division $UQ = D_8(X) - c$; the base holds 8 abscissae and the relation uses 5. It meets the letter of the "$m \geq 5$ cell" criterion without showing RR beating T in a cell with a tower; round 7′s m5 t4 had RR 25 to 60 times wider than T. (2) At m3 t5 RR's widest (31,911) exceeds T's lower bound (23,852), and round 7′s completed T had 23,720: the width advantage most likely reverses at $t = 5$. (3) T is not the lab's strongest $m = 3$ system: round 7′s arm F has widest 1,224 (m3 t4) and 10,493 (m3 t5); against F, RR is a trade (different curves and memory, so not combined into a claim). (4) Round 7′s T ran at 6 GiB; with sound bounds a cap can only leave a cell undecided. Full review: `followups/red-team-p1-p2.md`.

4. Next steps

Rerun T at m5 t3 with at least 6 GiB, then RR and T at m5 t4 on a host with at least 16 GiB; add round 7′s arm F at m3 t3–t5 on the same instances. The Dickson remainder tower's faithfulness is Lean-checked as `dickson_tower_faithful` (aburan28/crypto-lean#3) [pcc:unverified].

5. Round 2, card A: capped cells completed and arm F

Protocol `PROTOCOL-R2.md` (frozen before data, #3693); card `CARD-p1-r2-a` (executed as specified); 26 jobs at 6 GiB, the two T jobs at m5 t3 capped again. Analysis `p1r2analyze` (Rust) with the bound reading frozen before data. [pcc:unverified]

cell	RR	T	F	RR vs T	RR vs F
m3 t3	D 4, 632	D 5, 883	D 5, 231	RR dominates	trade
m3 t4	D 4, 2,801	D 6, 7,129	D 5, 1,224	RR dominates	trade
m3 t5	D 5, 31,911	D 6, 23,978	D 6, 10,493	trade	trade
m5 t3 (no tower)	D 5, 2,878	D ≥ 8, ≥ 82,927	n/a	RR below (bounds)	n/a

Control met exactly (T on the Kummer curve at m3 t5: D 6, widest 23,720, round 7′s values). Against the lab's strongest $m = 3$ system F, RR is a trade in all 9 cells: one degree lower, 2.3 to 3.0 times wider. Against T, RR dominates at t3 and t4 and is a trade at t5. The m = 5 cell with a tower (card B, T24 m5 t4, 13 GiB) decides rule 3.

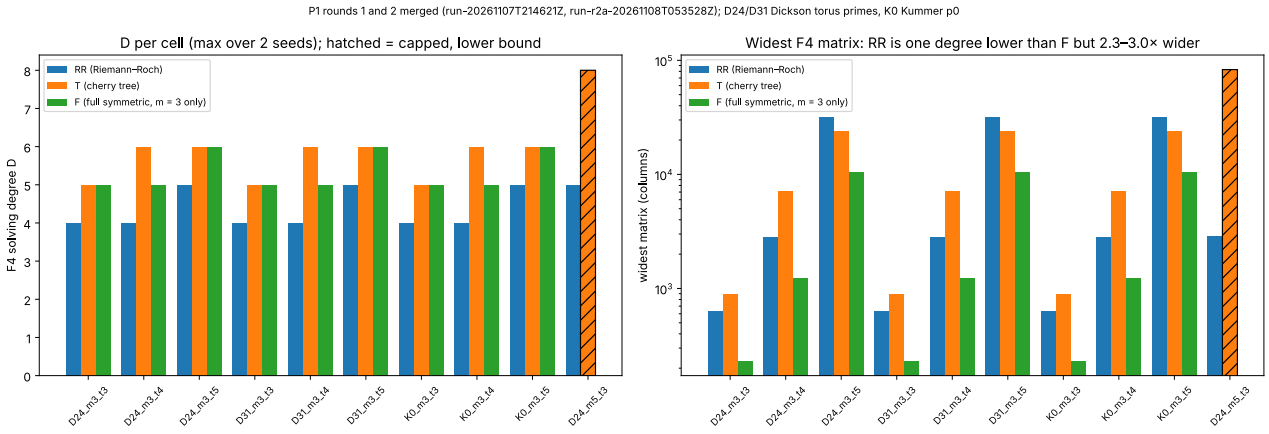


Figure 2: RR, T and F per cell, rounds 1 and 2 merged; hatched bars are capped runs (lower bounds).

6. Round 2, card B: the first m = 5 cell with a tower

Card `CARD-p1-r2-b`, T24 m5 t4, seed 2, 13 GiB and 10,800 s, alone on the host. RR timed out after 10,800 s at degree ≥ 6 (working in degree 7), widest ≥ 146,277; T hit the 13 GiB cap after 2,619 s at degree ≥ 9, widest ≥ 662,876. **Rule 3: undecided under the frozen text** (RR did not complete) [pcc:unverified]. Descriptively RR was at a lower degree with 4.5 times narrower matrices when stopped, limited by time not memory. Deciding it needs RR run to completion on a larger host with about a day's time limit.

7. Round 3: T24 m5 t4 on Iso Lab EKS, 56–60 GiB and 24 h (PROTOCOL-R3)

Same instrument, instance (seed 2; each pod re-emitted the system and its sha256 matched round 2′s) and frozen rule 3; only the limits and the host changed (RR `ulimit -v` 60 GiB, T 56 GiB, `timeout` 86,400 s, one msolve process per Spot pod). Analysis `analysis/p1r3analyze` ⟶ `analysis/derived_r3/decision_r3.json`.

attempt	host	outcome	D	widest
T a1	m7i.4xlarge Spot	lost: Spot interruption after 1 h 30 min	—	—
T a2	m7i.4xlarge Spot	**cap at 56 GiB after 21,267 s**, degree-9 step	**≥ 9**	**≥ 1,305,345**
RR a1	r7a.4xlarge Spot	lost: Spot interruption after 15 min	—	—
RR a2	r7a.4xlarge Spot	lost: Spot interruption after 2 h 33 min (after 10 h Pending)	—	—
RR a3	r6a.4xlarge Spot, pinned	lost: Spot interruption after 52 min	—	—

Bounds after rounds 2 and 3: RR D ≥ 6, widest ≥ 146,277 (R2b); T D ≥ 9, widest ≥ 1,305,345 (R3). **Rule 3: still undecided** — no RR attempt completed, and the frozen reading decides only for a completed RR. Every loss is a CloudTrail `BidEvictedEvent`; partial records are kept and supply no bound. Descriptively, all RR attempts stopped inside the same degree-7 step (matrix 464,562 × 390,983, never finished); T's degree-8 step completed at 262,201 columns. The obstacle is the Spot pool, not the computation: four interruptions in five pods. Cheapest next step: RR alone on a host that is not reclaimed (On-Demand or a qualified resumable route).

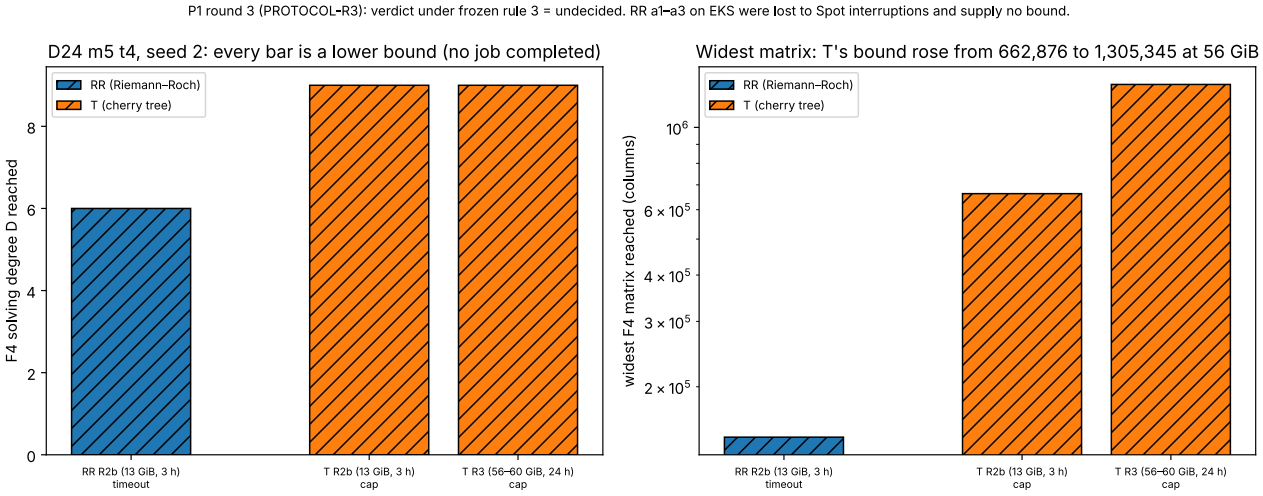


Figure 3: T24 m5 t4, seed 2, rounds 2b and 3: every bar is a lower bound (hatched); T's widest bound doubled at 56 GiB, RR never got past its degree-7 step on Spot.